

Ilyan EL HAJJAJI

Engineering Student in Electronics & Automation (3EA) *Toulouse INP-ENSEEIHT*

Amiens / Toulouse, France • ilyan.el-hajjaji@etu.inp-n7.fr • ePortfolio • LinkedIn

Profile

First-year engineering student with a solid background in mathematics, physics, and computer science. Driven by **semiconductors** and **automation**. Seeking an international exchange and an engineering internship.

Target Mobility

National Taiwan University
Semiconductor ecosystem

EPFL (Switzerland)
Microtechnique & precision control

TU Munich (Germany)
Automation & embedded systems

Technical Skills

Control
PID tuning, Hall-effect sensors.

Software & Tools
MATLAB, Simulink, Python, C, OCaml, Git/GitHub, LaTeX.

Languages

- **French:** Native
- **English:** Fluent (B2+)
- **Spanish:** Intermediate (B2)
- **German / Arabic / Mandarin:** Beginner

Interests

- **Football:** 4 years club practice (teamwork, communication).
- **Travel:** Intercultural discovery.

Engineering Projects

Active Feedback Magnetic Levitation System

- Built an analog closed-loop circuit to keep a metal ball in stable levitation at a fixed point without physical contact.
- Designed a real-time PID controller using Hall-effect feedback to dynamically reject perturbations.

Automotive ABS Optimization Model

- Simulated and then tested 3 different types of braking to optimize the ABS.
- Designed and benchmarked 3 control strategies: **relay control**, **classical PID**, and **pseudo-AI optimized PID**.
- Achieved encouraging results with the latest optimized model.

Autonomous Obstacle-Avoidance Mobile Robot

- Built an autonomous robotic car navigating structured indoor environments.
- Implemented a real-time wall-avoidance reactive pathfinding algorithm using IR sensors.

Education

Toulouse INP - ENSEEIHT *2026 – Present*
M.Sc. in Electrical, Electronics & Automation (3EA) *Toulouse, France*

- Training in electronics, signal processing, control engineering, and the modeling of dynamic systems.

Lycée Pierre Corneille *2023 – 2026*
CPGE MPI (Maths, Physics, Computer Science) *Rouen, France*

- Intensive nationwide preparation for competitive Grande École engineering exams.

Lycée Sainte-Famille d'Amiens *2020 – 2023*
French Baccalaureate (Maths & Physics-Chemistry) – Highest Honors *Amiens, France*

Research & Conference Exposure

- **IEEE International Conference (Seoul, 2023):** Attended sessions on advanced semiconductors and automation.
- **MIS Laboratory (Amiens):** Shadowed researchers working on non-linear control algorithms and robotics testbeds.